

Segmentation models

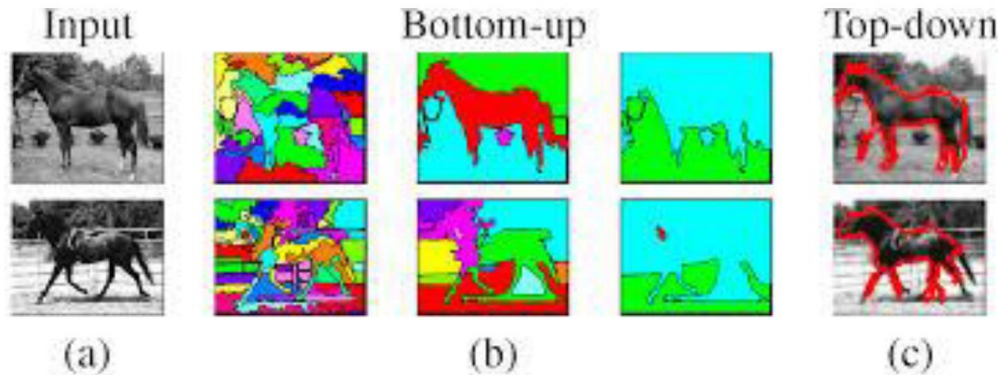
- Types of segmentation models
 - Semantic, instance, panoptic
- Major segmentation architectures
 - CNN, U-net
 - Transformer-based
 - VLM-based
 - Hybrid
- Segmentation foundational model (input-wise)
 - SAM
- Medical imaging-specific architectures
 - MedSAM, Instruction-tuned segmentation models
 - Custom architectures

The different notions of segmentation

- Computer vision
 - Image partitioning into objects and background
- Medical imaging
 - Anatomy segmentation
 - Anomaly segmentation

Bottom-up?
Or top-down?

Supervised?
Or unsupervised?



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Semantic segmentation

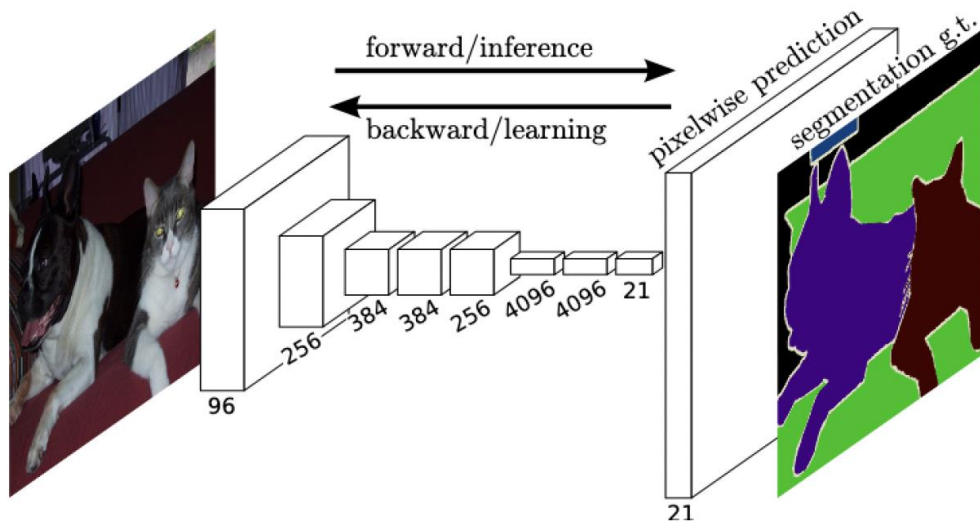


- Labels each pixel by class
- E.g. FCNet, DeepLab, PSPNet

Semantic segmentation example: FCN

- Fully-supervised method
- Used in many other networks for the segmentation head
- Using purely convolutional setup

1. Feature extraction (Encoder)
Edges, shapes, organs
2. Upsampling/decoding
transposed convolutions or
interpolations
3. 1x1 convolution
4. Pixel-wise classification



<https://arxiv.org/abs/1411.4038>

**Fully Convolutional Networks for
Semantic Segmentation**

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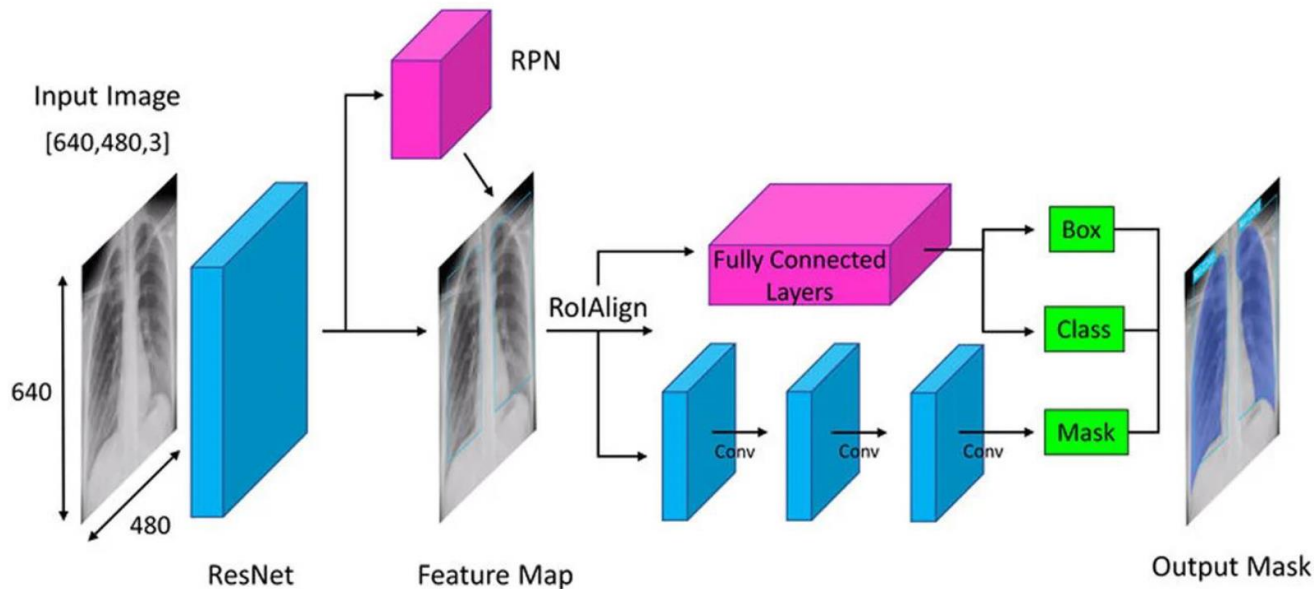
Instance segmentation



- Each region has a distinct label
- Combine object detection and semantic segmentation
- E.g. Mask R-CNN, Yolo-8

Instance segmentation example: Mask RCNN

- Up to 1500 class labels can be detected and labeled



<https://arxiv.org/abs/1703.06870>, K.He et al. Mask R-CNN

>48,000 citations

The different notions of segmentation

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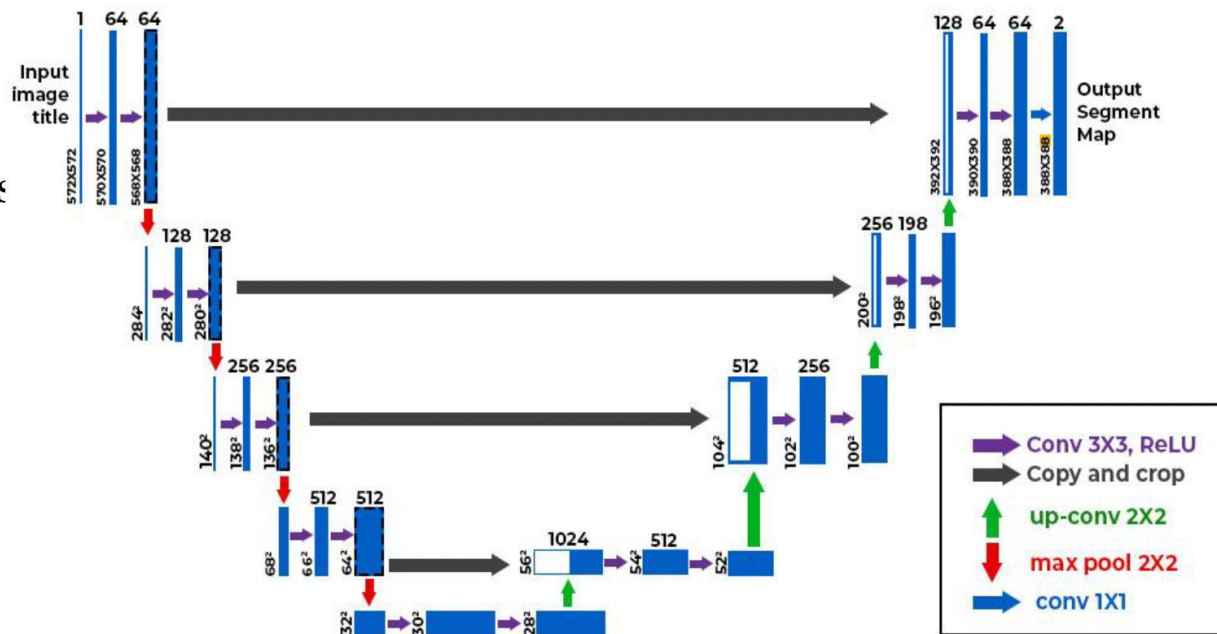
panoptic segmentation



- Labels each pixel by class and also identifies different instances of the same class
- E.g. Efficient PS, Panoptic DeepLab, RS-DINO, HyperDETR, DinoV2, FCN+MaskRCN

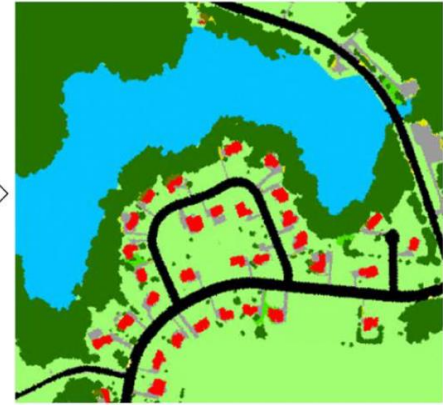
U-net - A classic CNN-style segmentation model

- Extensively used in medical imaging
- Works for both anatomy and anomaly detection
- Top-down segmentation works best for medical images since intensities are intermingled.
- Works well when the pattern to learn is in a fixed place
- Skip connections help remember features
- Fully CNN-based



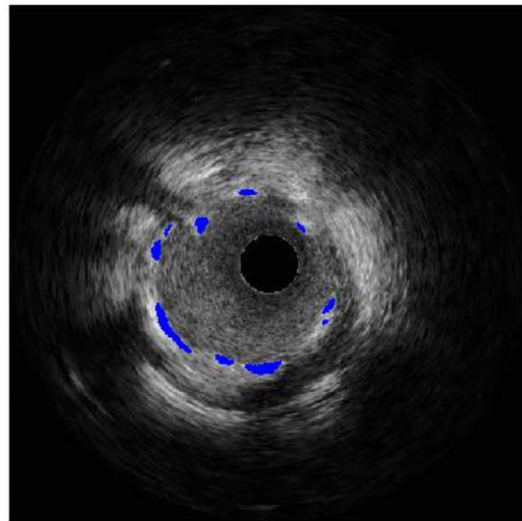
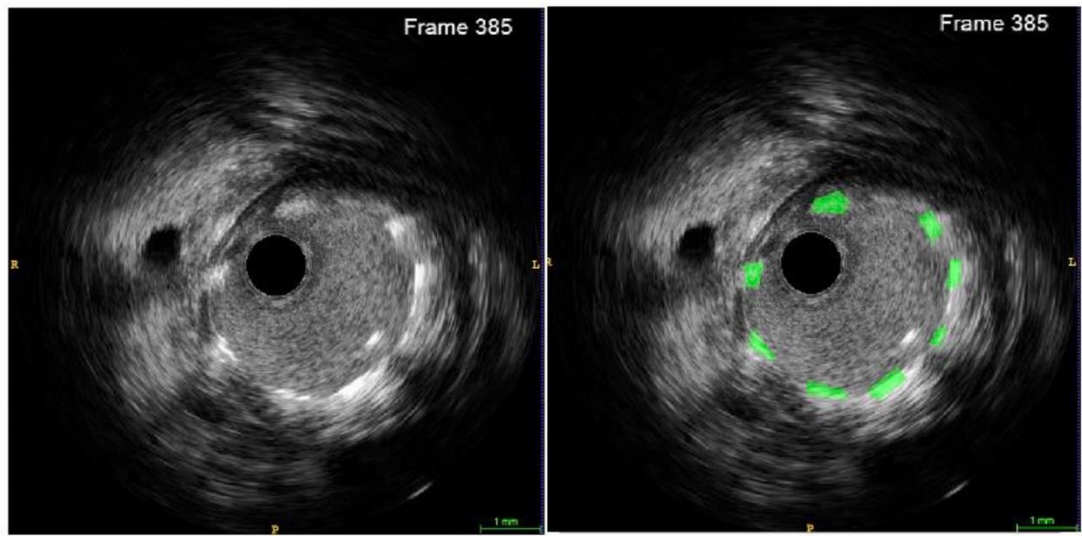
U-net applications

- Both medical and non-medical applications



U-Net issues

- Detections are not smooth
- Maintaining continuity is difficult
- Fragmentation can occur
- Small tumors and anatomical regions
- Stable contour detection



Basic architectures for segmentation FM

- Based on two major architectures:
 - CNN
 - Transformers
- Different combinations of encoders and decoders
 - U-net : Convolutional encoder and decoder
 - TransU-net: Transformer encoder and CNN decoder
 - SETR: Vision transformer and CNN decoder
 - Segformer: transformer encoder and decoder
 - CLIPSeg: CLIP encoder and transformer decoder
- Go from limited object labels to open vocabulary segmentation
- Mostly still supervised, but newer unsupervised open set models also coming back

Design choices for encoders and decoders for segmentation

1. **Encoder choice** depends on:

- Input modality (2D vs 3D)
- Need for global context (transformers excel)
- Pretraining availability

2. **Decoder choice** depends on:

- Desired output resolution
- Computational budget
- Multi-scale vs single-scale features

3. **Skip connections** are critical to:

- Recover fine-grained spatial details
- Preserve edges, organ boundaries, or object contours

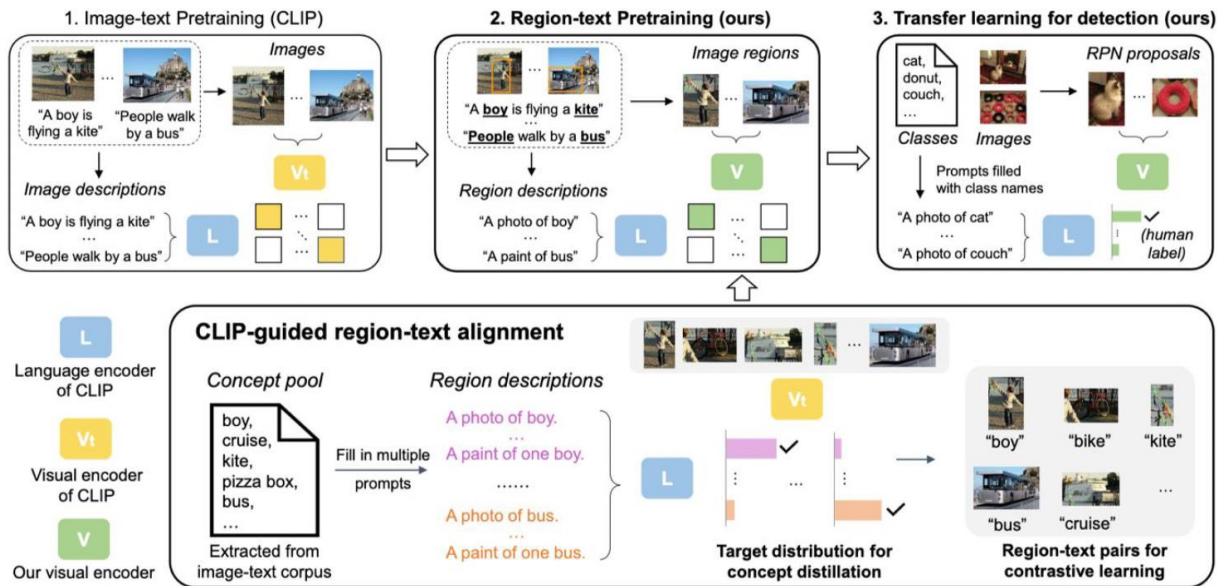
Combinations of encoder and decoders

Model	Encoder	Decoder	Notes
U-Net	Custom CNN (Conv blocks)	Symmetric upsampling + skip connections	Widely used in medical imaging
UNet++	CNN	Nested skip connections	Better feature fusion
SegNet	VGG16	Upsampling with pooling indices	Lightweight, efficient
DeepLabV3+	ResNet / Xception	Atrous Spatial Pyramid Pooling (ASPP) + upsampling	Multi-scale context
FPN (Feature Pyramid Network)	ResNet / DenseNet	Top-down pyramid merging	Often used for detection + segmentation
SegFormer	MiT (Transformer encoder)	Lightweight MLP-based decoder	High performance with attention + low compute
Mask2Former	Swin Transformer / ResNet	Transformer decoder with queries	Supports panoptic, instance, semantic segmentation
Attention U-Net	CNN	Skip connections + attention gates	Focuses decoder on relevant encoder features
3D U-Net / V-Net	3D CNN	3D upsampling + skip connections	Volumetric medical image segmentation

Segmentation models derived from VLM models

- Use joint visual-textual information to aid segmentation
- Use CLIP underneath for the encoding
- Built separate decoders to aid in segmentation
 - CNN-based
 - Transformed-based decoders
- Segmentation at the level of bounding boxes
 - RegionCLIP
 - VILD
 - GLIP
- Segmentation as full regions
 - CLIPSeg

VLM – Visual Grounding

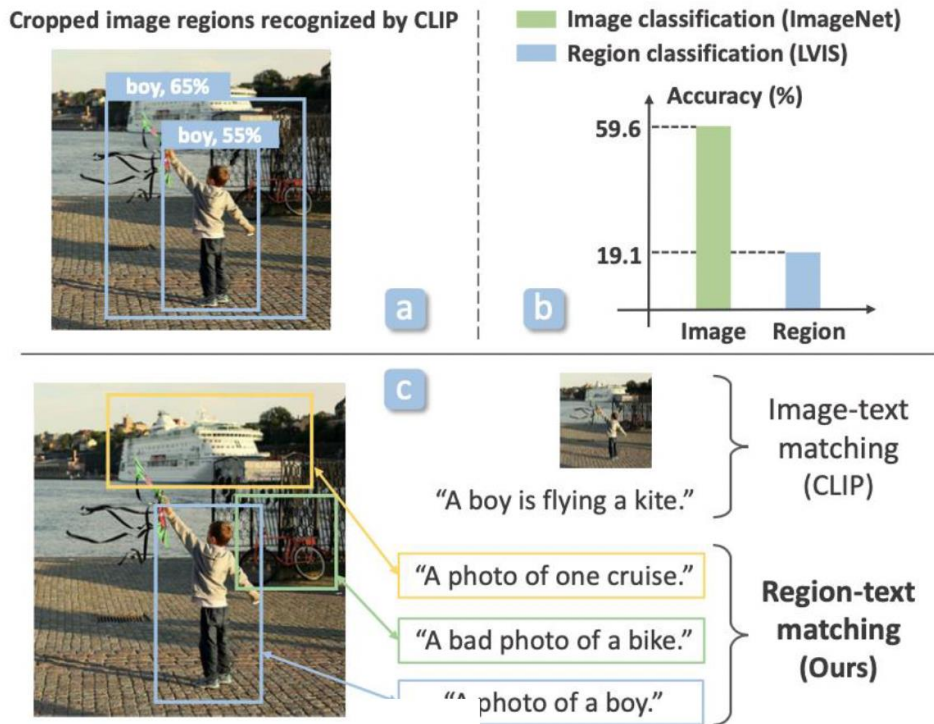


RegionCLIP – Contrastively learned region-text alignment

Using VLM for generalized instance segmentation

- Use region crops from RPN
- Initial CLIP image-to-text to label
- What if we do region detection and CLIP on regions fails to recognize objects

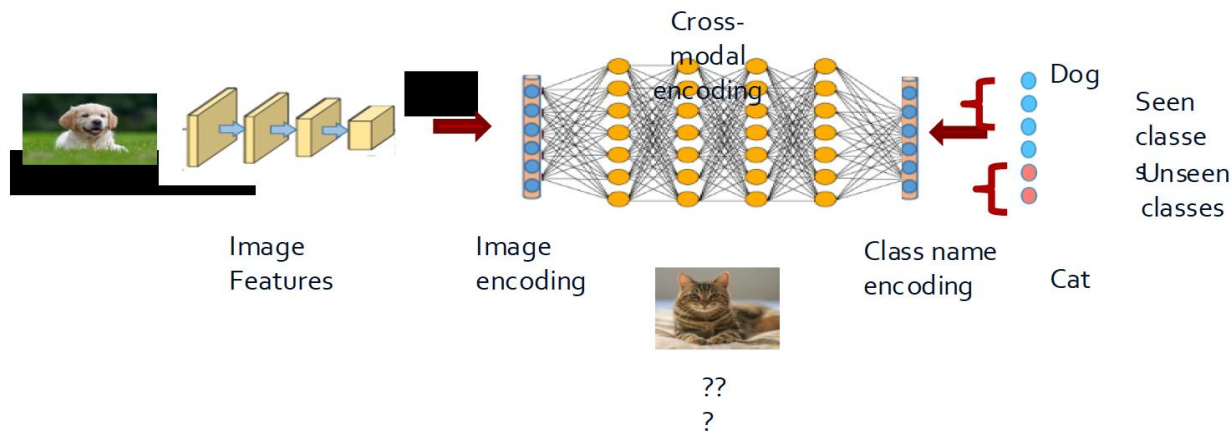
The labels should be known at inference time



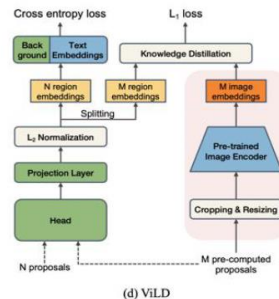
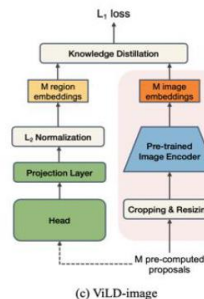
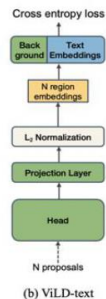
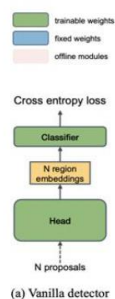
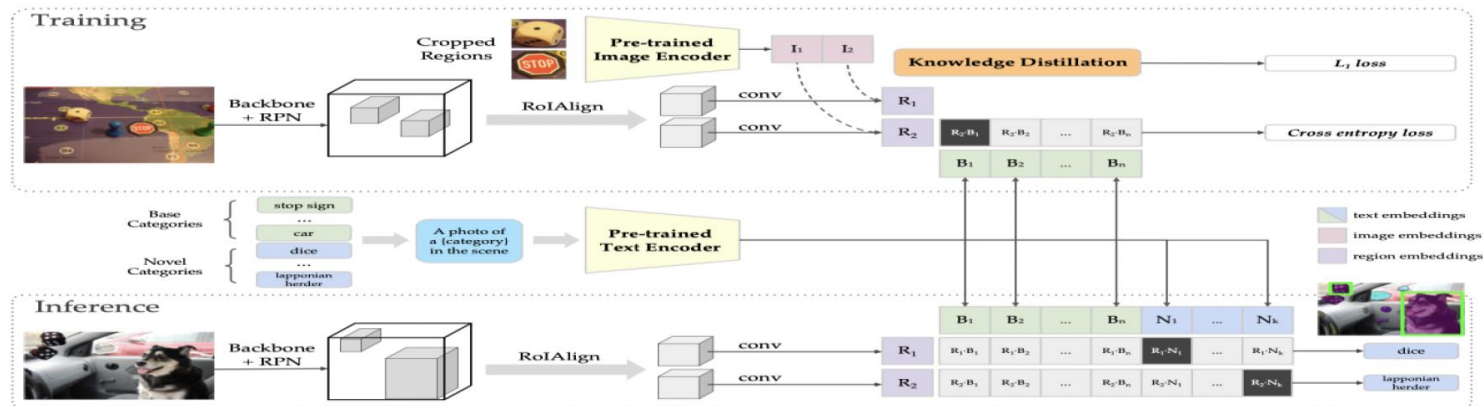
RegionCLIP – Contrastively learned region-text alignment

VILD architecture

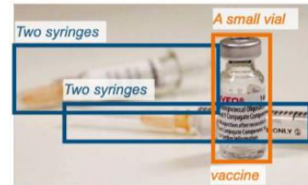
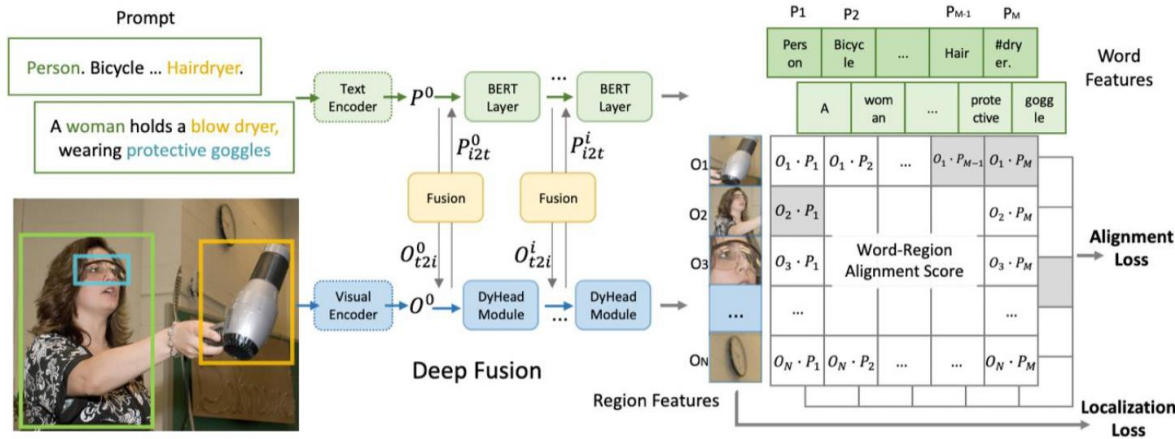
- Combines pre-trained FM (CLIP) with MaskRCNN-style RPN and zero-shot learning methods
- Distill the knowledge from the alignment of region embedding to image embeddings of cropped regions during training.



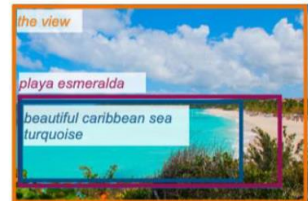
ViLD – Visual Grounding



VLM – Phrase Grounding



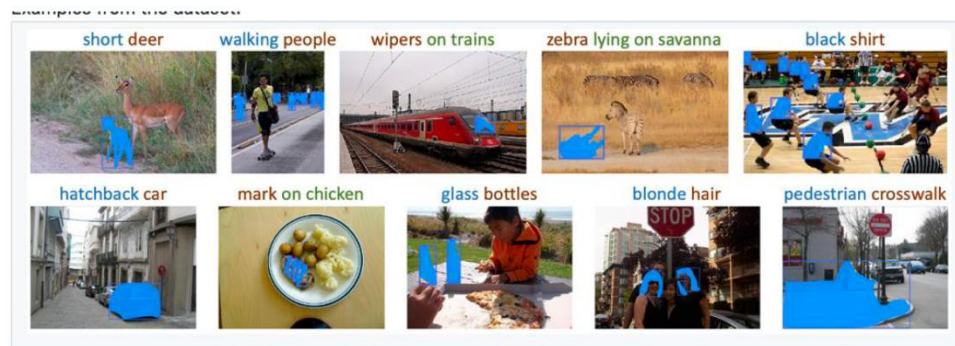
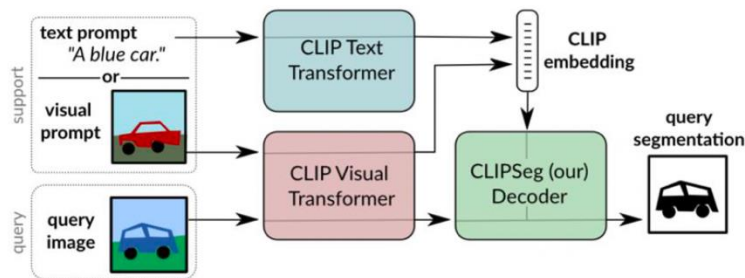
Two syringes and a small vial of vaccine.



playa esmeralda in holguin, cuba. the view from the top of the beach. beautiful caribbean sea turquoise

- Given an image and a corresponding caption, the **Phrase Grounding** task aims to ground each entity mentioned by a noun phrase in the caption to a region in the image.
- GLIP – grounded language pre-training

CLIPSeg

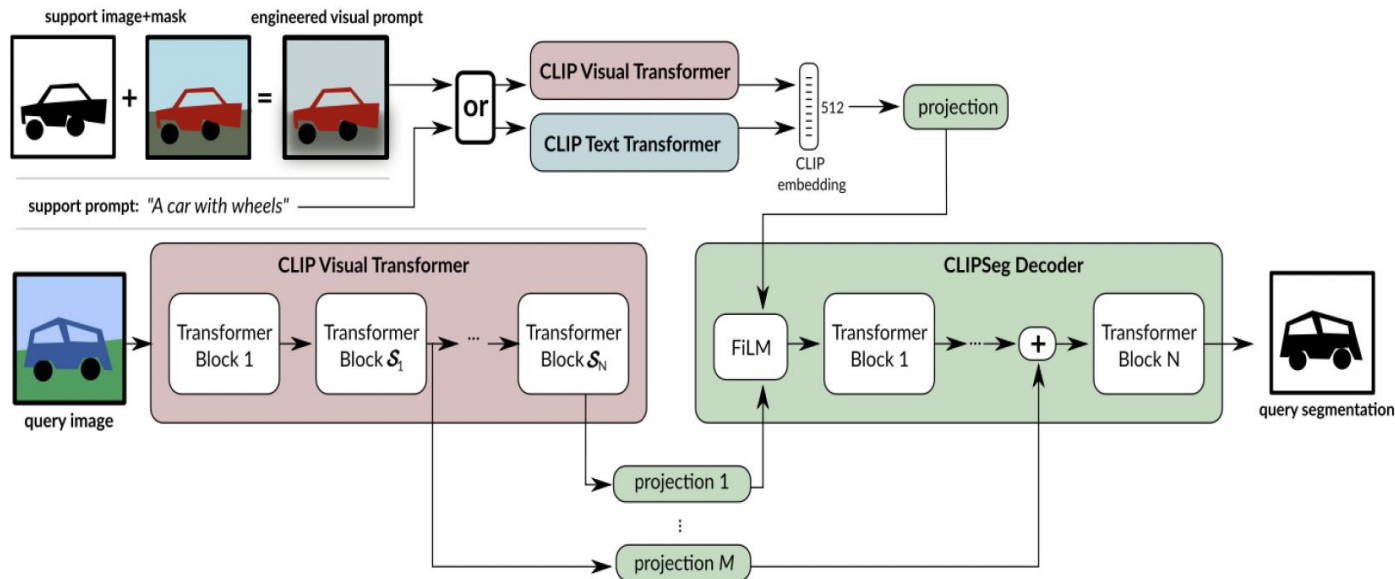


- Uses the PhrasCut dataset of regional masks derived from VisualGenome
- Combined CLIP with a lightweight decoder

CLIPSeg: Image segmentation using text and image prompts

ClipSeg

- Skip connections to preserve the visual context from 3 of the layers of the ViT to the decoder as in U-net
- The decoder incoming dimension is 64

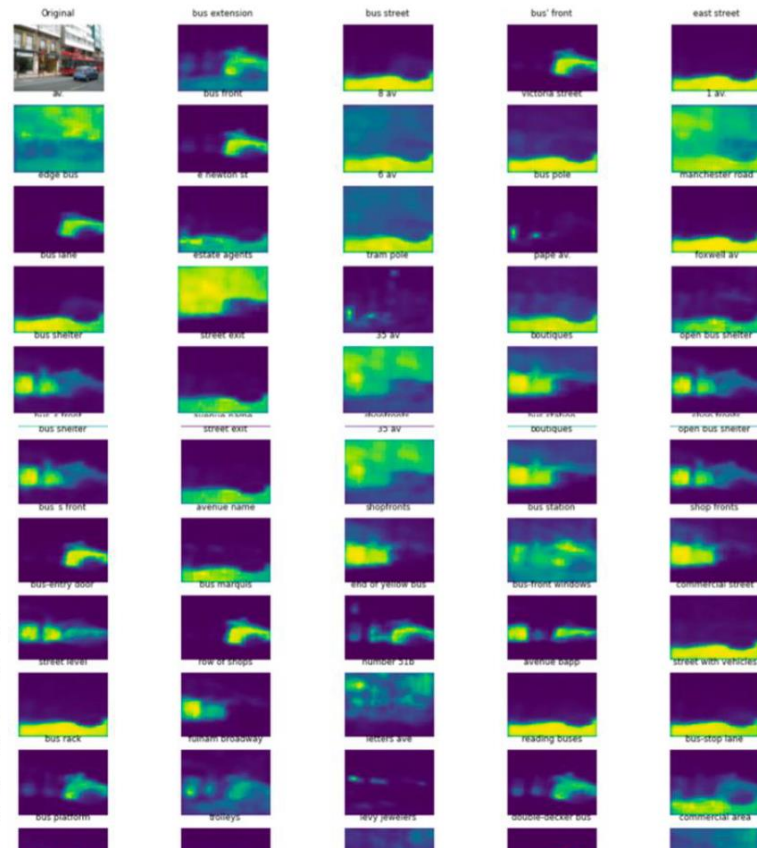


CLIPSeg results

(a)



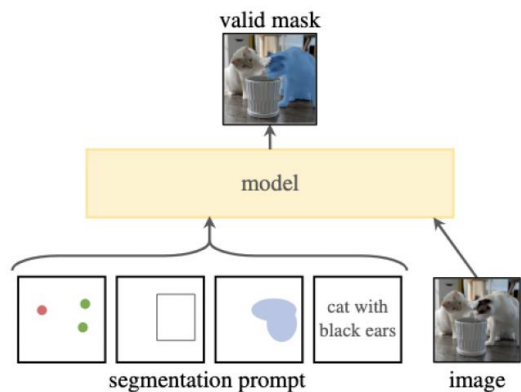
(b)



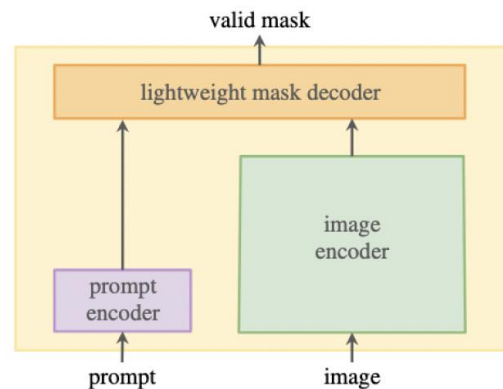
'victoria street', 'manchester road', 'street level', 'foxwell av', '6 av', 'east street', 'street with vehicles', 'bus-entry door', 'bus's front', 'shopfronts', 'commercial street', 'bus rack', 'bus front', 'street exit', '8 av', 'estate agents', 'bus lane', '35 av', 'avenue name', 'bus marquis', 'trolleys', "bus' front", 'avenue bapp', 'row of shops', 'boutiques', 'e newton st', 'bus street', 'bus station', 'bus pole', 'bus lane', 'bus rack', 'bus platform', 'busways', 'ivy jewellers', 'scooter-becker bus', 'commercial area', 'bus-stop lane', 'reading buses', 'avenue bapp', 'bus station', 'bus front windows', 'end of yellow bus', 'shopfronts', '33 av', '35 av', 'tram pole', '6 av', '8 av', 'bus street', 'bus extension', 'Original'

Segment Anything (SAM)

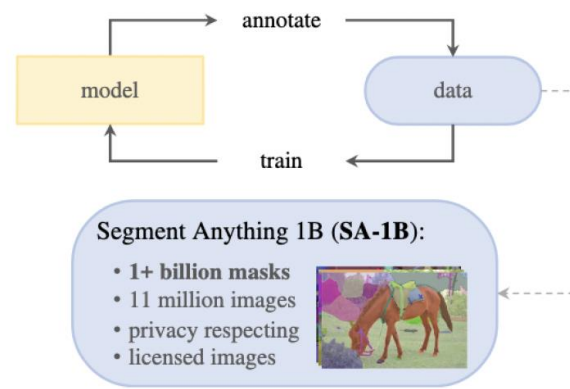
- A foundation model for image segmentation.
- A promptable model and pre-train it on a broad dataset using a task that enables powerful generalization



(a) **Task:** promptable segmentation



(b) **Model:** Segment Anything Model (SAM)



(c) **Data:** data engine (top) & dataset (bottom)

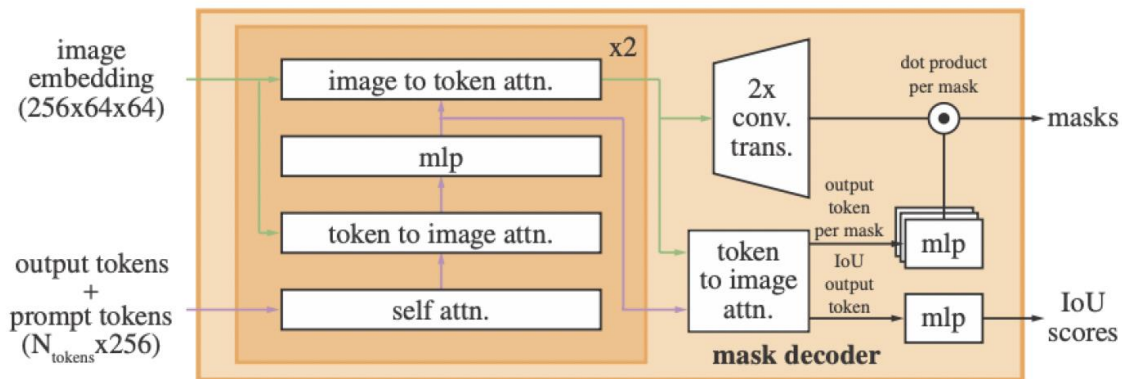
Different kinds of prompts to aid in segmentation

SAM components

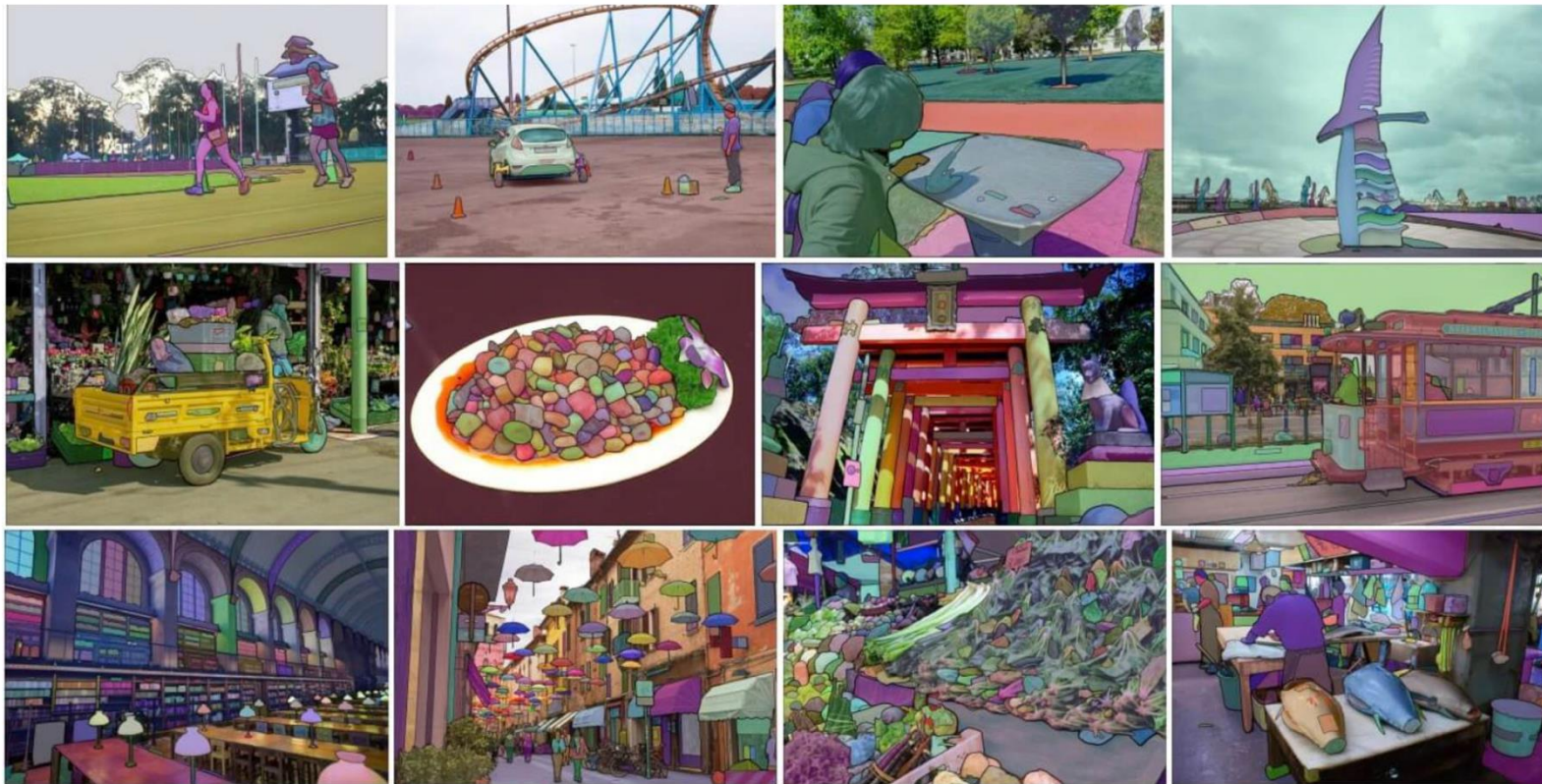
- Image Encoder:
 - Encoder portion of the Masked Auto-encoder which is a pre-trained Vision Transformer (ViT) with adaptations to process high resolution inputs,
 - input resolution of 1024×1024 obtained by rescaling the image and padding the shorter side. The image embedding is therefore $64 \times 64 \times 256$.
- Prompt encoder:
 - Produces a 256-dimensional vector
 - Points
 - Point is represented as a sum of positional encoding of the points's location and one of the two learned embeddings to indicate either a **foreground point** or **background point**.
 - Bounding boxes:
 - Boxes are represented by an **embedding pair of corners**.
 - Text:
 - CLIP text-encoding
 - Masks:
 - Downscaled versions of masked images CNN-style and flattened to 256-dimensional vectors

Segment Anything – Mask Decoder

- Image embeddings and prompt embeddings are mapped to the final mask
- uses prompt self-attention and cross-attention in two directions (prompt-to-image embedding and vice-versa) to update *all* embeddings.
- MLP maps the output token to a dynamic linear classifier, which then computes the mask foreground probability at each image location.

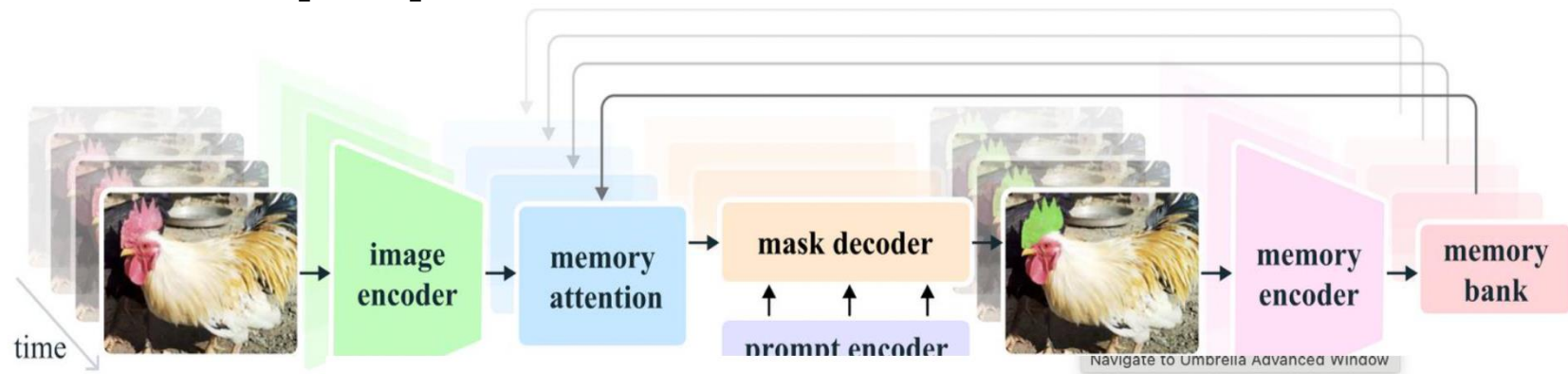


SAM results

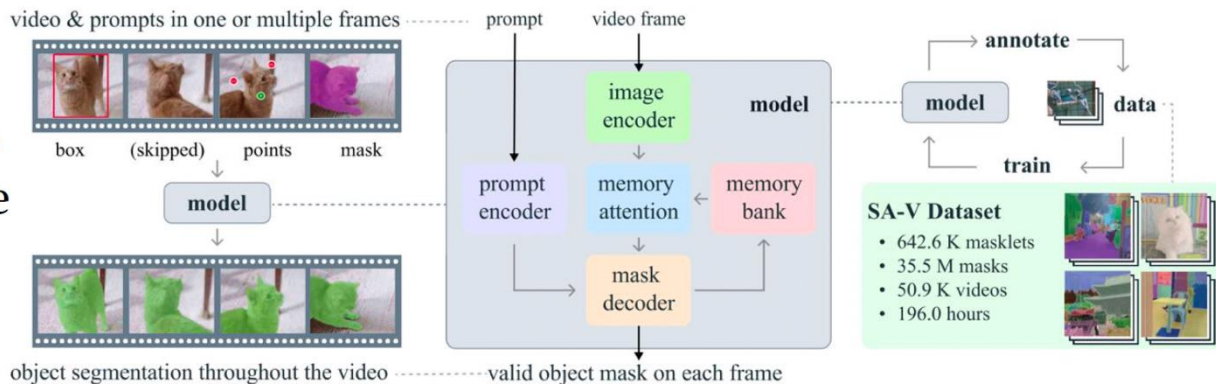


SAM-2 and SAM-3

SAM-3 can take text prompts



Extends to video sequences by retaining previous frame information



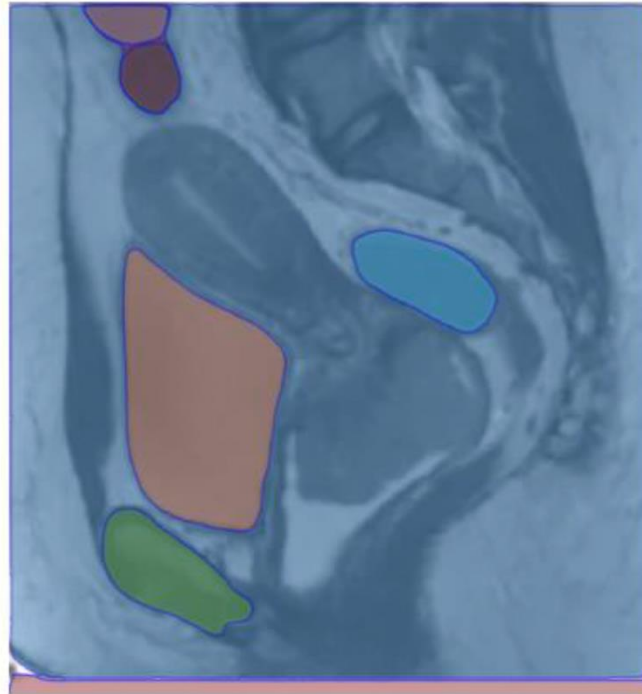
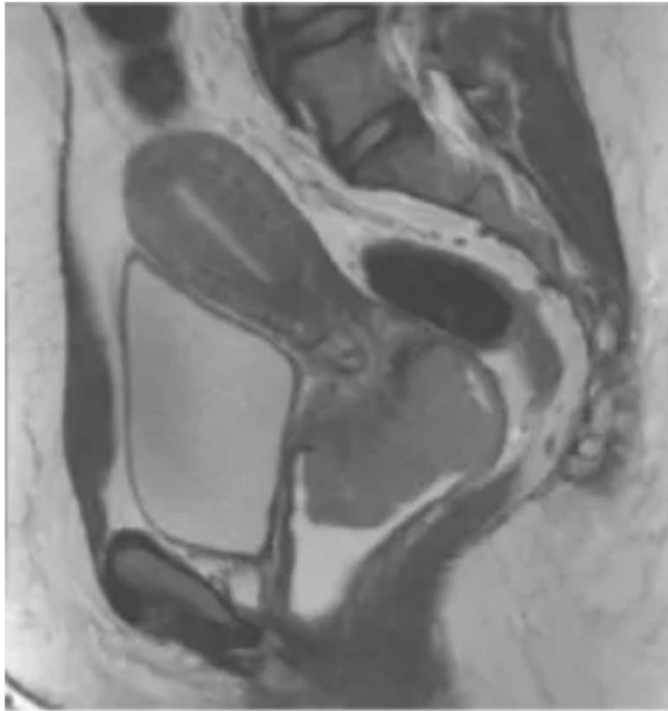
State of segmentation

- Automatic class-agnostic object segmentation
 - SAM in automatic mask generation
 - No recognition of objects though
- Fully automatic semantic segmentation
 - output pixel-wise class labels automatically (e.g. Mask2Former, DeepLabV3)
- Open vocabulary automatic segmentation
 - Automatically segment and label arbitrary objects without prompts. – an open research problem (still needs to know the vocabulary even if not seen an object)

There are three levels of "automatic":

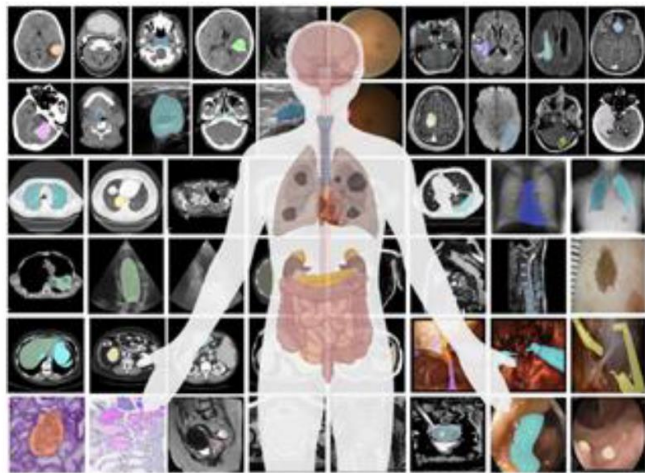
Type	Needs Prompt?	Needs Class List?	Open Vocabulary?
SAM automatic mode	✗	✗	✗ (no labels)
DeepLab-style models	✗	✓	✗
CLIPSeg-style	✓ (text)	✗	✓
GLIP-style	✓ (text)	✗	✓
Research hybrids	Sometimes	Sometimes	Partially

How well does SAM do on medical images?



Foundational models for segmentation

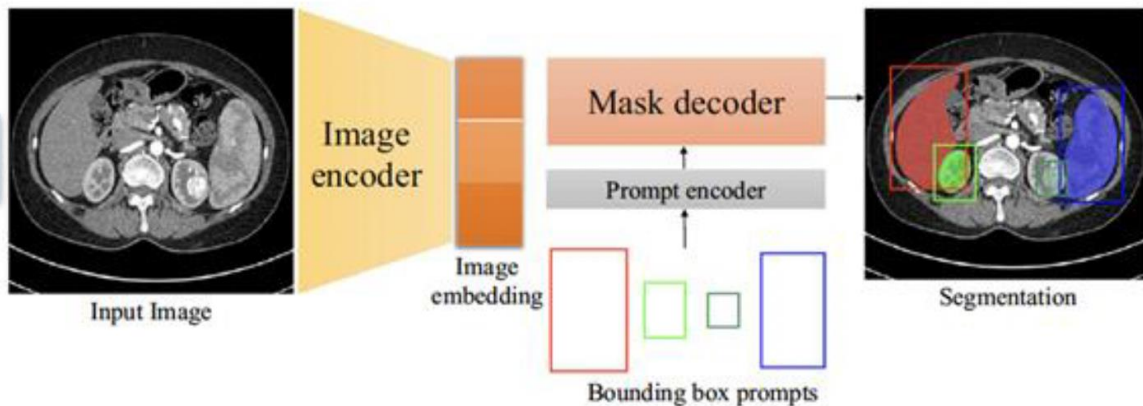
MedSAM



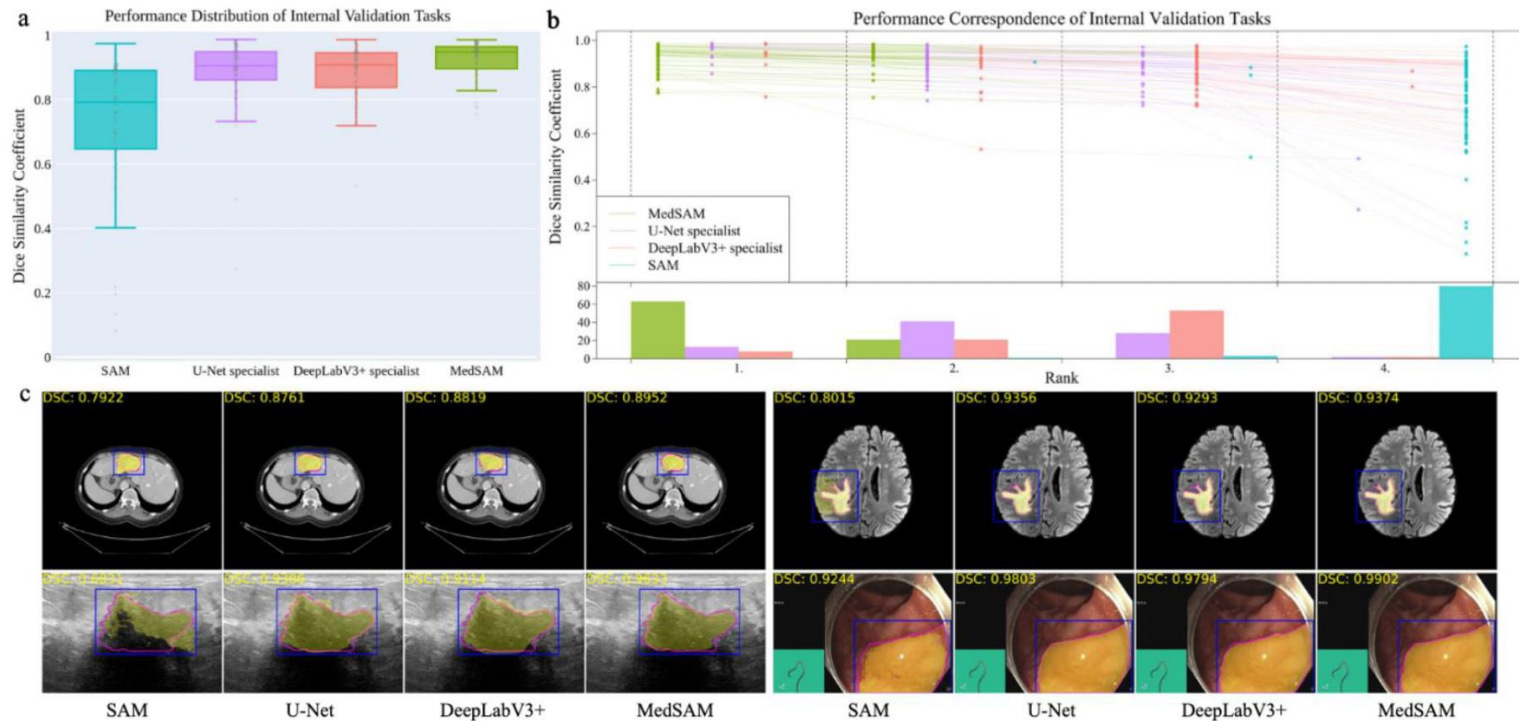
Trained on an huge amount of data

- 10 imaging modalities
- 30 cancer types
- Over 1.5M mask-image pairs

Pipeline

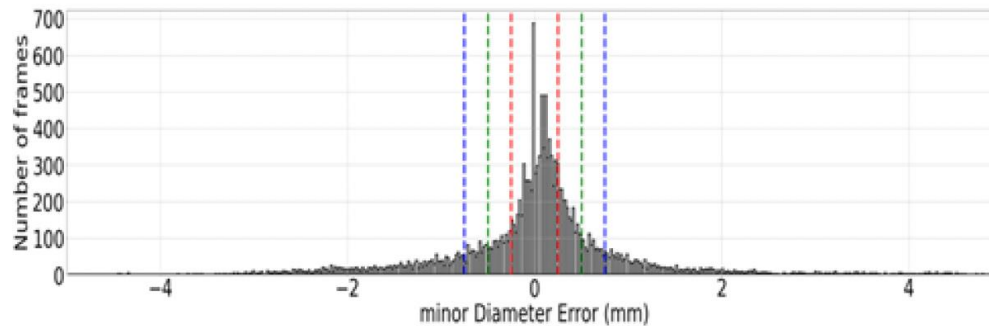
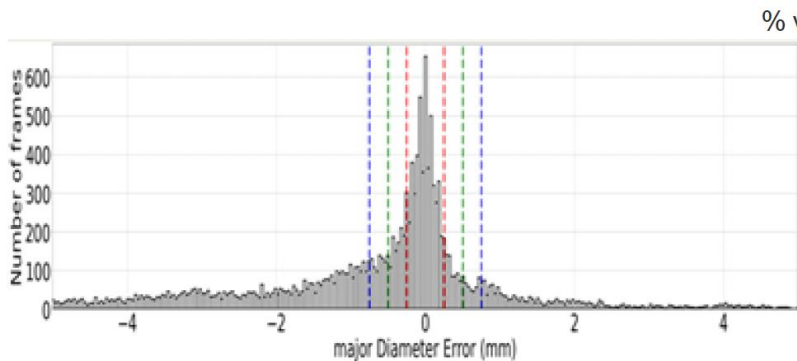
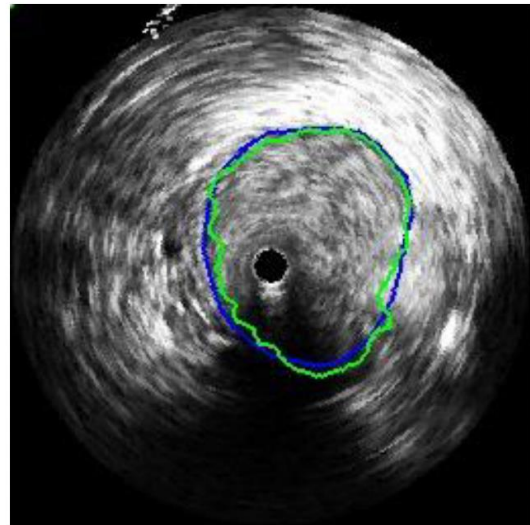


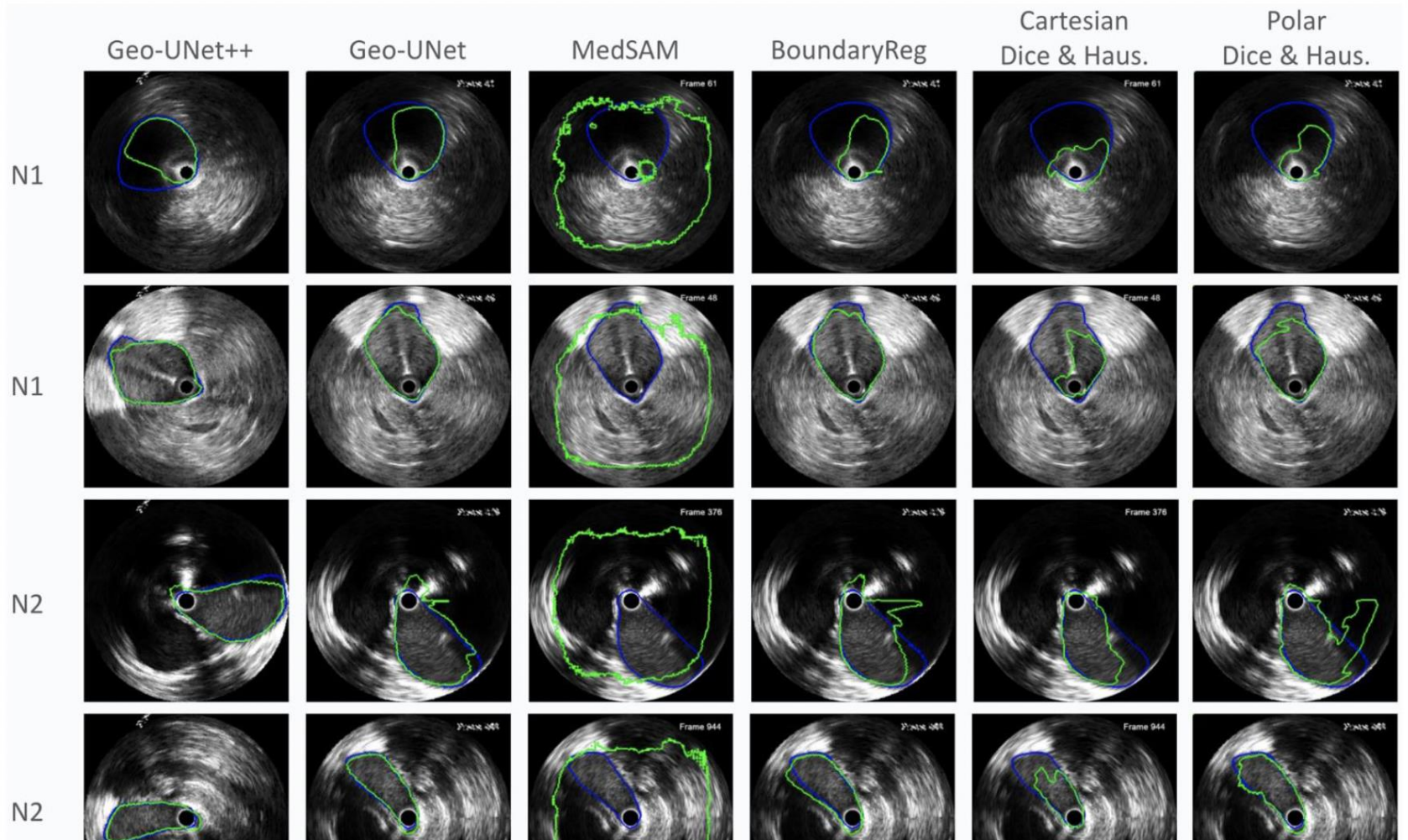
MedSAM generalization



Are FMs ready for high-precision segmentation?

- Needed for accurate sizing of the stents
- Based on normal frames identification
- Estimates the maximum and minimum diameter
- Major and minor diameter errors within:
 - 0.25/0.5/0.75mm for 50/90/95% of all N1 frames.
 - 0.5/0.75mm for 50/70% of frames for N2 frames
- N2 normal are mainly used for vessel compression detection and not stent sizing
- Simple U-net will not suffice as single regions cannot be ensured
- Contours may not closely follow the lumen boundaries





Results

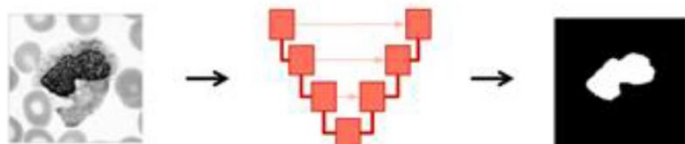
Methodology	Test Dice (avg/std)	% Frames w. Maj. Dia. err. within 0.25/0.50/0.75mm	% Frames w. Min. Dia. err. within 0.25/0.50/0.75mm
Against Baselines (N1 frames)			
Geo-UNet++	<u>0.95/0.045</u>	<u>66/84/90</u>	73/89/94
Geo-UNet	0.95/0.034	69/84/90	69/85/ <u>91</u>
MedSAM [10]	0.31/0.087	0/0/0	0/0/0
BoundaryReg [4]	0.94/0.043	60/78/86	<u>70/86/91</u>
Cart. Dice & Haus.	0.93/0.051	61/77/83	62/79/87
Polar Dice & Haus.	0.94/0.038	<u>66/80/87</u>	67/84/90
Against Baselines (N2 frames)			
Geo-UNet++	0.88/0.094	<u>41/59/69</u>	60/80/87
Geo-UNet	<u>0.87/0.10</u>	47/64/73	<u>57/76/85</u>
MedSAM [10]	0.23/0.085	0/0/0	0/0/0
BoundaryReg [4]	<u>0.87/0.093</u>	36/54/65	55/74/84
Cart. Dice & Haus.	<u>0.83/0.12</u>	32/44/52	44/63/74
Polar Dice & Haus.	<u>0.86/0.12</u>	40/58/ <u>69</u>	55/74/83
Against Ablations (N1 frames)			
Geo-UNet	0.95/0.034	69/84/90	69/85/91
w/o CDFeLU	0.94/0.035	69/82/88	<u>65/83/90</u>
w/o pixel-wise pred.	<u>0.95/0.039</u>	67/81/87	69/85/91
Against Ablations (N2 frames)			
Geo-UNet	<u>0.87/0.10</u>	47/64/73	57/76/85
w/o CDFeLU	<u>0.86/0.10</u>	45/63/ <u>72</u>	<u>53/71/81</u>
w/o pixel-wise pred.	0.88/0.092	46/62/71	57/76/85

Meta learning in UniverSeg

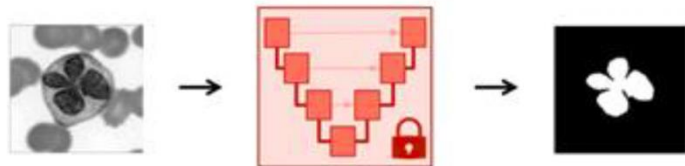
UniSeg

Traditional Approach

1. Design and train a task-specific model.

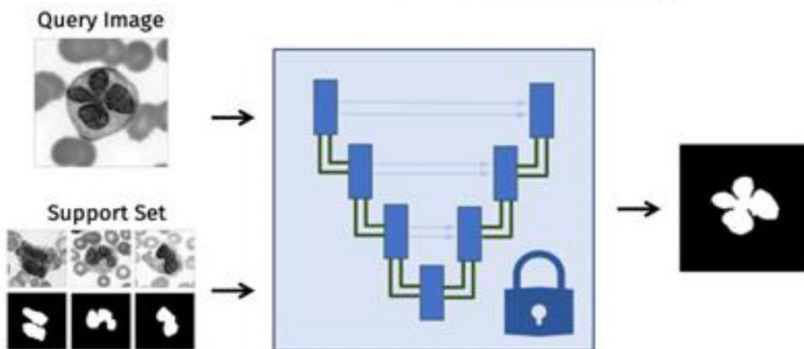


2. Predict new images with the trained model.

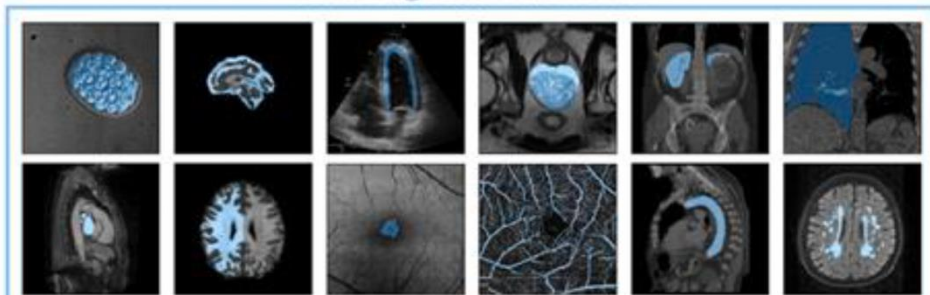


UniverSeg Approach

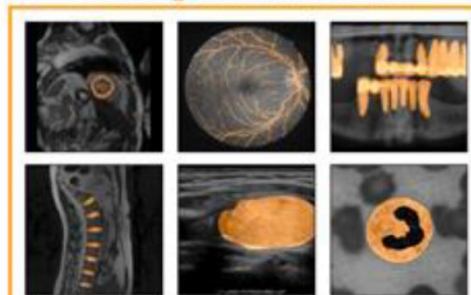
With a trained UniverSeg model, predict new images for the new task from a few labeled pairs without retraining.



Train Segmentation Tasks



Test Segmentation Tasks

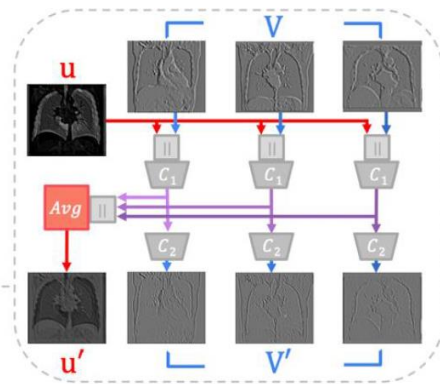
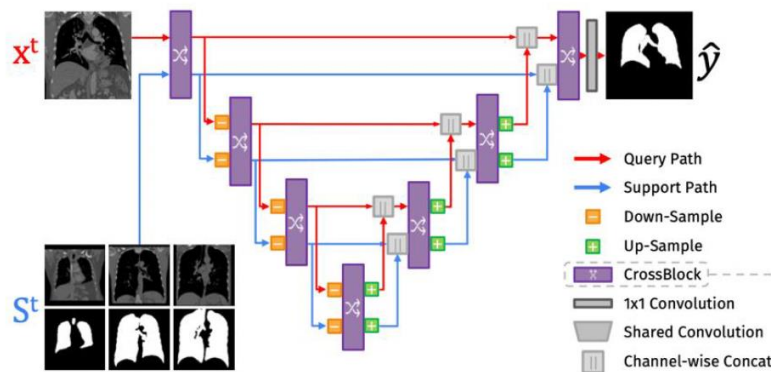


Instructional tuning of segmentation models

- Based on U-net style convolutional architecture
- Uses multi-scale cross-block features between instructional labeling sets and given image
- No retraining needed or fine-tuning needed!
- Trained on Megamedical dataset: 53 datasets, 23 medical domains, 16 modalities

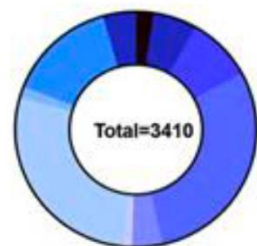
Cross-convlayer

- Convolve the query with each support
- Reduces them together to incorporate information about what to segment based on the support labels
- The result is updated representations for both query and support



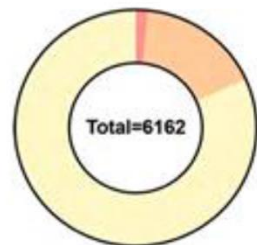
Butai et al. Universal medical image segmentation, ICCV'2023

Combining U-net and CLIP for anatomy and tumor segmentation



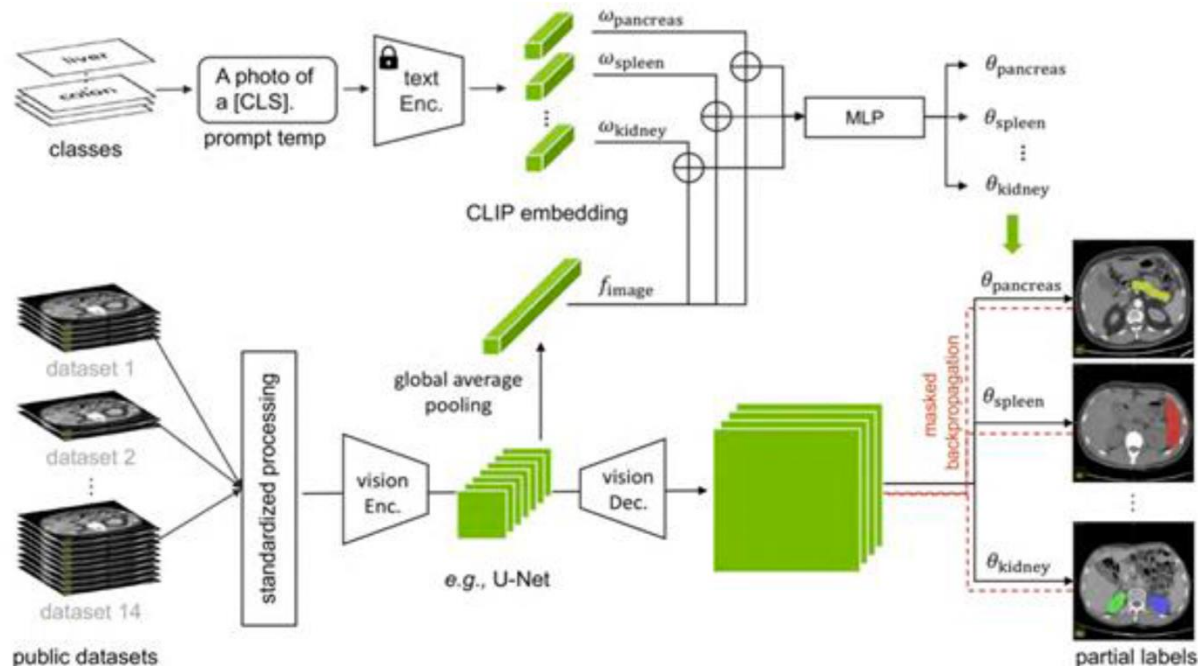
for training

- 82 Pancreas-CT (1;0)
- 201 LITS (1;1)
- 300 KITS (1;1)
- 1000 AbdomenCT-1K (4;0)
- 140 CT-ORG (4;0)
- 40 CHAOS (4;0)
- 947 MSD (7;4)
- 50 BTCV (13;0)
- 500 AMOS (15;0)
- 150 WORD (16;0)



for testing

- 100 3D-IRCADb (13;0)
- 1024 TotalSegmentator (104;0)
- 5038 JHH (21;0)



Slides from MICCAI 2024 tutorial

Foundational models for segmentation

CLIP-Driven

Main idea

Text branch

(generates text embedding for class k) $\mathbf{w}_k$

Visual branch-encoder

(generates visual embedding for image x) $\mathbf{f}$

Text-based controller MLP
(generates class parameters)

$$\theta_k = MLP(\mathbf{w}_k \oplus \mathbf{f})$$

$$\theta_k = \{\theta_{k_1}, \theta_{k_2}, \theta_{k_3}\}$$

Visual branch-decoder

(generates visual embedding for image x)

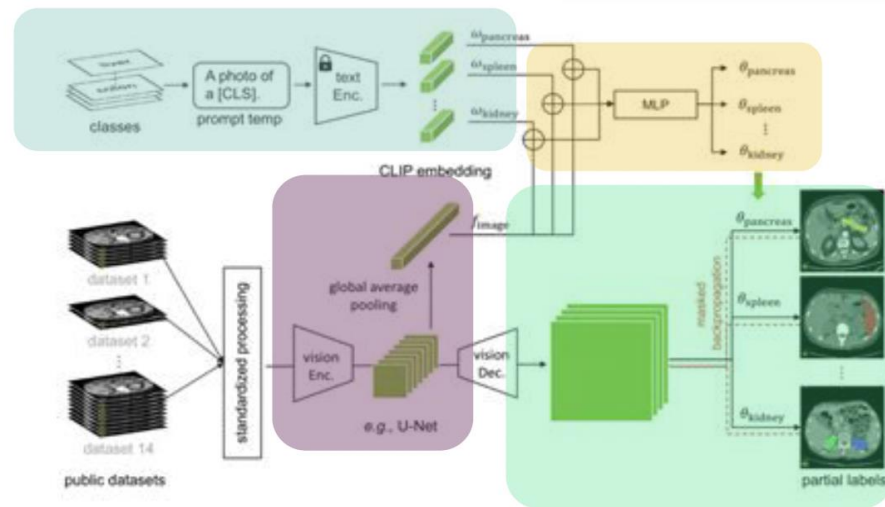
$$\mathbf{P}_k = \text{sigmoid}(((\mathbf{F} * \theta_{k_1}) * \theta_{k_2}) * \theta_{k_3})$$

It represents foreground class k vs background

Training loss

Binary cross-entropy per class (and terms masked for those classes not present)

$$\mathcal{L} = \sum_{k=1}^K \mathbf{1}_{\{k \in y\}} \cdot \text{BCE}_k$$



Segmentation models

Model / Framework	Type	Strength
SAM 3	General Vision Foundation Model	Open-vocabulary & multi-instance segmentation (images + video)
SAM 2	Vision Foundation Model	Interactive prompt + video temporal consistency
SAM 1	Vision Foundation Model	Zero-shot general segmentation
ENSAM	Medical/3D segmentation variant	Efficient 3D segmentation under limited data
SAMPO	Enhanced prompt adaptation research	Intent-aware segmentation
TopoLoRA-SAM	Adaptation framework	Efficient fine-tuning for specific tasks
SAM2-UNeXT	High-resolution foundation adaptation	Stronger features across domains